

Junior Backend Developer Training Plan

Tech Stack

- Node.js (Express.js)
 - MongoDB
 - Elasticsearch
 - Redis (Basics)
 - BullMQ (Background Jobs & Scheduled Jobs)
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Phase 1: JavaScript & Node.js Fundamentals (Weeks 1-2)

Week 1: Modern JavaScript

Topics

- ES6+ Features
- Arrays & Objects
- Promises
- Async/Await
- Error Handling

Assignment

Build a simple CLI application with CRUD operations using JSON files.

Outcome

Understand asynchronous programming and modern JavaScript syntax.

Week 2: Node.js Fundamentals

Topics

- Event Loop
- Non-blocking I/O
- Modules
- NPM
- Environment Variables
- Logging
- File System Operations

Assignment

Build a REST API without Express.

Outcome

Understand how Node.js works under the hood.

Phase 2: Express.js Backend Development (Weeks 3-4)

Week 3: Express.js Basics

Topics

- Routing
- Middleware
- Request Lifecycle
- Error Handling
- Validation

Assignment

Build a Task Management API.

Outcome

Understand REST API development.

Week 4: Advanced Express.js

Topics

- Authentication
- Authorization
- JWT
- RBAC
- API Versioning
- Swagger/OpenAPI

Assignment

Implement Login, Registration, Roles, and Protected Endpoints.

Outcome

Build secure APIs.

Phase 3: MongoDB (Weeks 5-6)

Week 5: MongoDB Fundamentals

Topics

- Documents & Collections
- CRUD Operations
- Indexes
- Aggregation Basics

Assignment

Migrate the Task API to MongoDB.

Outcome

Understand document database concepts.

Week 6: Advanced MongoDB

Topics

- Schema Design
- References vs Embedding
- Aggregation Pipeline
- Transactions
- Performance Optimization

Assignment

Build a Mini Recruitment System.

Outcome

Design scalable MongoDB schemas.

Phase 4: Redis Fundamentals (Week 7)

Topics

- Redis Basics
- Key-Value Storage
- Expiration & TTL
- Counters
- Caching

Assignment

Implement:

- Session Cache
- API Cache
- Job Details Cache

Outcome

Understand caching strategies and cache invalidation.

Phase 5: BullMQ (Weeks 8-9)

Week 8: Queue Fundamentals

Topics

- Jobs
- Workers
- Delayed Jobs
- Retries
- Failure Handling

Assignment

Implement email sending through a queue.

Outcome

Understand asynchronous processing.

Week 9: Advanced BullMQ

Topics

- Repeated Jobs
- Scheduled Jobs
- Queue Monitoring
- Scaling Workers

Assignment

Create daily reports and reminder notifications.

Outcome

Operate production-grade background processing.

Phase 6: Elasticsearch (Weeks 10-11)

Week 10: Elasticsearch Fundamentals

Topics

- Indices
- Documents
- Mappings
- Search APIs
- Analyzers

Assignment

Implement Job and Candidate Search.

Outcome

Understand search engine fundamentals.

Week 11: Advanced Elasticsearch

Topics

- Full-Text Search
- Relevance Scoring
- Aggregations
- Multi-field Search
- Synonyms

Assignment

Build a Job Search API with filters and faceted search.

Outcome

Build production-grade search experiences.

Phase 7: Real-World Backend Engineering (Week 12)

Topics

Architecture

- MVC
- Repository Pattern
- Service Layer Pattern

Quality

- Unit Testing
- Integration Testing
- Code Reviews

Observability

- Structured Logging
- Error Tracking
- Monitoring

Security

- OWASP Basics
 - Input Validation
 - Rate Limiting
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Final Project

Build a Recruitment Platform Backend containing:

Features

- Authentication & Authorization
- Job Management
- Candidate Management
- Applications

Infrastructure

- MongoDB for persistence
- Redis for caching and rate limiting
- BullMQ for background jobs
- Elasticsearch for search

Additional Requirements

- Swagger Documentation
- Unit Tests
- Docker Setup
- Logging
- Validation
- Pagination

Expected Skills

By the end of the program, the developer should be able to:

- Build REST APIs with Express.js
- Design MongoDB schemas and aggregations
- Implement Redis caching strategies
- Build background jobs with BullMQ
- Create Elasticsearch indexes and search APIs
- Write unit and integration tests
- Apply security and performance best practices
- Understand production backend architecture